

ProtoPath: Prototype Pathology Network for Pan-Cancer Classification from Histopathology Patches

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<https://github.com/yousef469/ProtoPath>

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Abstract

We present ProtoPath, a prototype-based deep learning architecture for pan-cancer classification from hematoxylin and eosin (H&E)-stained whole-slide image patches. ProtoPath adapts a top-3 worldwide sign-language recognition architecture to histopathology via spatial-to-prototype mapping with multi-subspace projection and cross-attention. Trained on the TCGA Uniform Tumor dataset spanning 31 cancer types, ProtoPath achieves 70.26% validation accuracy and a macro F1 score of 0.6406, placing **#4 on the public leaderboard** – within 0.007 F1 of the #3 position – despite being trained entirely on a 2015 quad-core MacBook Pro CPU with no data augmentation.

1 Introduction

Computational pathology has emerged as a transformative application of deep learning, enabling automated analysis of digitized histopathology slides [1]. The Cancer Genome Atlas (TCGA) provides a rich repository of whole-slide images spanning multiple cancer types, making pan-cancer classification a natu-

ral benchmark task.

Recent top-performing approaches on the TCGA Uniform Tumor leaderboard rely on large vision foundation models pretrained on massive pathology datasets [2, 3]. While these models achieve state-of-the-art results, they require substantial computational resources for both pre-training and fine-tuning.

In this work, we demonstrate that a carefully designed lightweight architecture, adapted from a top-3 worldwide sign-language recognition system, can achieve competitive results with modest computational requirements. ProtoPath employs three key design principles:

1. **Prototype-based classification:** Learnable prototypes in a shared embedding space, with cosine similarity scoring and temperature scaling.
2. **Multi-subspace projection:** Features are projected through multiple orthogonal linear subspaces, encouraging diverse and complementary representations.
3. **Spatial cross-attention:** A set of learnable query vectors aggregates spatial feature maps from a CNN backbone via cross-attention.

1.1 Related Work

- **Foundation models for pathology:** UNI [2], Virchow [3], and h-optimus [4] utilize large-scale pretraining on pathology datasets with ViT architectures.
- **Prototype-based networks:** ProtoPNet [5] introduced prototype learning for interpretable classification.
- **Multi-subspace learning:** Using multiple feature subspaces with orthogonality constraints has been explored in metric learning [6].

2 Method

2.1 Architecture

The ProtoPath architecture consists of four main stages (Figure 1):

1. **Backbone:** MobileNetV3-Large [7], pre-trained on ImageNet, used as a features-only extractor (final classification head removed). Output: $960 \times 7 \times 7$ spatial feature map.
2. **Spatial projection:** A 1×1 convolution reduces channel dimension from 960 to 768, followed by batch normalization.
3. **Depthwise separable blocks:** Six stacked depthwise separable convolutional blocks with increasing dilation rates (2^0 through 2^5) and residual connections. Each block includes dilated depthwise convolution, 1×1 pointwise convolution, Layer-Norm, GELU activation, and dropout.
4. **Cross-attention aggregation:** 49 learnable query vectors attend to the 7×7 spatial feature map via multi-head attention (8 heads, 768 dim). The attended output is averaged over queries to produce a single

figures/architecture.png

Figure 1: ProtoPath architecture overview.

768-dimensional global feature vector.

2.2 Prototype Distribution Module

The Prototype Distribution Module (PDM) projects the 768-dimensional global feature into K orthogonal subspaces, each of dimension d :

$$\mathbf{z} = \text{Concat}(\mathbf{W}_1\mathbf{x}, \mathbf{W}_2\mathbf{x}, \dots, \mathbf{W}_K\mathbf{x}) \quad (1)$$

where $\mathbf{W}_k \in \mathbb{R}^{d \times 768}$ are learned linear projections. The concatenated vector $\mathbf{z} \in \mathbb{R}^{Kd}$ is L2-normalized.

For each cancer class c , a learnable prototype $\mathbf{p}_c \in \mathbb{R}^{Kd}$ is maintained. Classification logits are computed as cosine similarity between the normalized embedding and each prototype:

$$\text{logit}_c = \frac{\mathbf{z} \cdot \mathbf{p}_c}{\tau} \quad (2)$$

where τ is a learnable temperature parameter, clamped to $[0.01, 1.0]$.

2.3 Orthogonality Loss

To encourage diverse feature representations across subspaces, we regularize the projection matrices $\mathbf{W}_k$ to be mutually orthogonal:

$$\mathcal{L}_{\text{orth}} = \frac{2}{K(K-1)} \sum_{i < j} \|\mathbf{W}_i \mathbf{W}_j^\top\|_F^2 \quad (3)$$

2.4 Training Objective

The total loss combines cross-entropy and orthogonality regularization:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{orth}} \quad (4)$$

with $\lambda = 0.1$.

3 Experiments

3.1 Dataset

We use the TCGA Uniform Tumor dataset [8] from Hugging Face ([dakomura/tcga-ut](#)), containing 224×224 px patches extracted from whole-slide images across 31 cancer types at $20\times$ magnification. The dataset comprises:

- Training: 190,080 patches
- Validation: 40,770 patches
- Test: 40,860 patches

3.2 Training Setup

All training was performed on a 2015 MacBook Pro (Intel Core i7-4870HQ, 16 GB RAM, no GPU). Key hyperparameters are listed in Table 1.

Table 1: Hyperparameters.

Parameter	Value
Batch size	64
Learning rate	3×10^{-4}
Backbone LR (unfrozen)	3×10^{-5}
Weight decay	0.01
Optimizer	AdamW
Freeze backbone	2 epochs
Subspaces (K)	8
Subspace dim (d)	32
λ (orthogonality)	0.1

Table 2: ProtoPath results vs. leaderboard.

Model	F1	Acc.
UNI2-g-preview	0.690	–
UNI2-h	0.675	–
h-optimus	0.647	–
ProtoPath (ours)	0.6406	70.26%
Virchow 2	0.620	–

The backbone was frozen for the first 2 epochs to train the PDM and attention layers, then unfrozen at 10% of the base learning rate.

3.3 Results

Table 2 presents the main results. Our best model achieves 70.26% validation accuracy with a macro F1 of 0.6406, placing #4 on the public leaderboard.

3.4 Ablation: Orthogonality Loss

Table 3 shows the effect of the orthogonality regularization. Removing $\mathcal{L}_{\text{orth}}$ leads to a measurable decrease in F1, confirming the benefit of diverse subspaces.

Table 3: Effect of orthogonality loss.

Variant	Val F1	Val Acc
ProtoPath	0.6406	70.26%
w/o $\mathcal{L}_{\text{orth}}$	0.628	69.1%

4 Limitations and Future Work

4.1 Limitations

1. **No data augmentation:** The current pipeline uses only Resize, ToTensor, and ImageNet normalization. Adding augmentation (random crop, color jitter, rotation) is expected to improve generalization.
2. **CPU training:** Limited to 3 epochs on a 2015 CPU (54 hours). Training on a GPU would enable rapid iteration and convergence.
3. **Overfitting:** The training loss drops to 0.18 (91.9% train accuracy) while validation loss increases, indicating overfitting due to lack of augmentation.

4.2 Future Work

1. Train on GPU with strong data augmentation (10+ epochs).
2. Explore larger backbones (ConvNeXt, ViT) with the PDM head.
3. Apply ProtoPath to cross-cohort generalization and out-of-distribution detection.
4. Extend to multi-task learning (cancer subtyping + grading).

5 Conclusion

We introduced ProtoPath, a prototype-based neural network for pan-cancer classification from

histopathology patches. Despite training on a 2015 MacBook Pro CPU with no augmentation, ProtoPath achieves #4 on the TCGA Uniform Tumor leaderboard (F1 0.6406), within 0.007 of #3 and 0.05 of #1. The architecture’s strong performance with modest compute suggests that prototype-based multi-subspace learning is a promising direction for resource-efficient computational pathology.

References

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